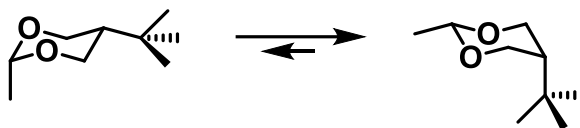


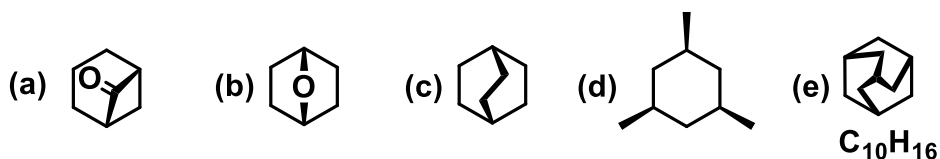
CHM102 Basic Organic Chemistry

Conformations

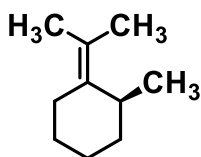
Q1. Cyclohexanes with bulky substituents in the equatorial position rather than the axial position are more stable. However, the following equilibrium shows the preference for bulky substituent in the axial position. What could be the reason?



Q2. Draw the proper conformer of the following compounds.

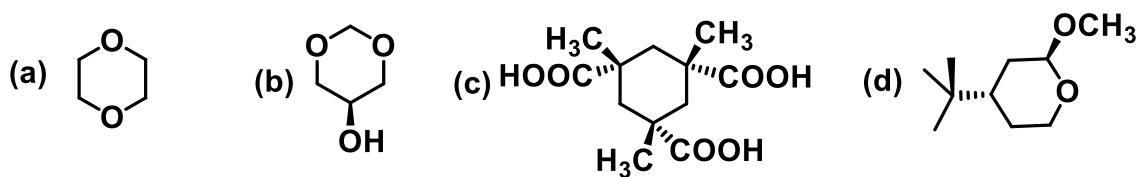


Q3. Draw the possible chair conformers of the following compounds. Which one of the chair forms is stable? Indicate the reason.

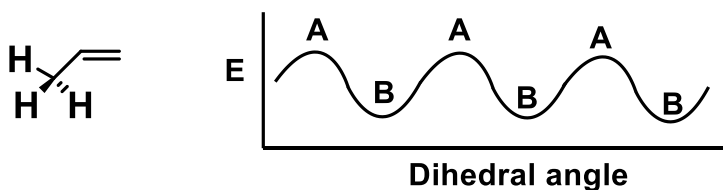


Q4. Indicate the possible conformers of hydrazine (NH_2NH_2), (fluoromethyl)amine (FCH_2NH_2) and fluoromethanol (FCH_2OH) using Newman projection and indicate the stability order with reasons.

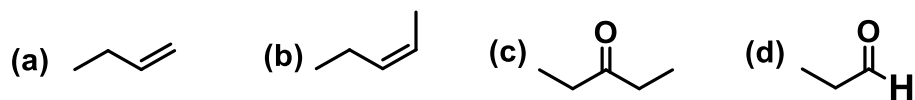
Q5. Draw the most stable conformations for the following compounds:



Q6. If the conformational changes can be represented for propene as follows, identify the conformations A and B and indicate the reason.



Q7. Identify the major conformers and indicate the most preferred conformations of the following compounds with reason:



Q8. Identify the conformational preferences in 1,3-butadiene and 3-buten-2-one.

Q9. Draw the Newman projections of planar and puckered cyclobutane and compare the possible strains in each one of them.

Q10. Draw the conformations of the following compounds.

